

Unit 3

DOING THE BEST YOU CAN: SCARCITY,
WELLBEING, AND WORKING HOURS

OUTLINE

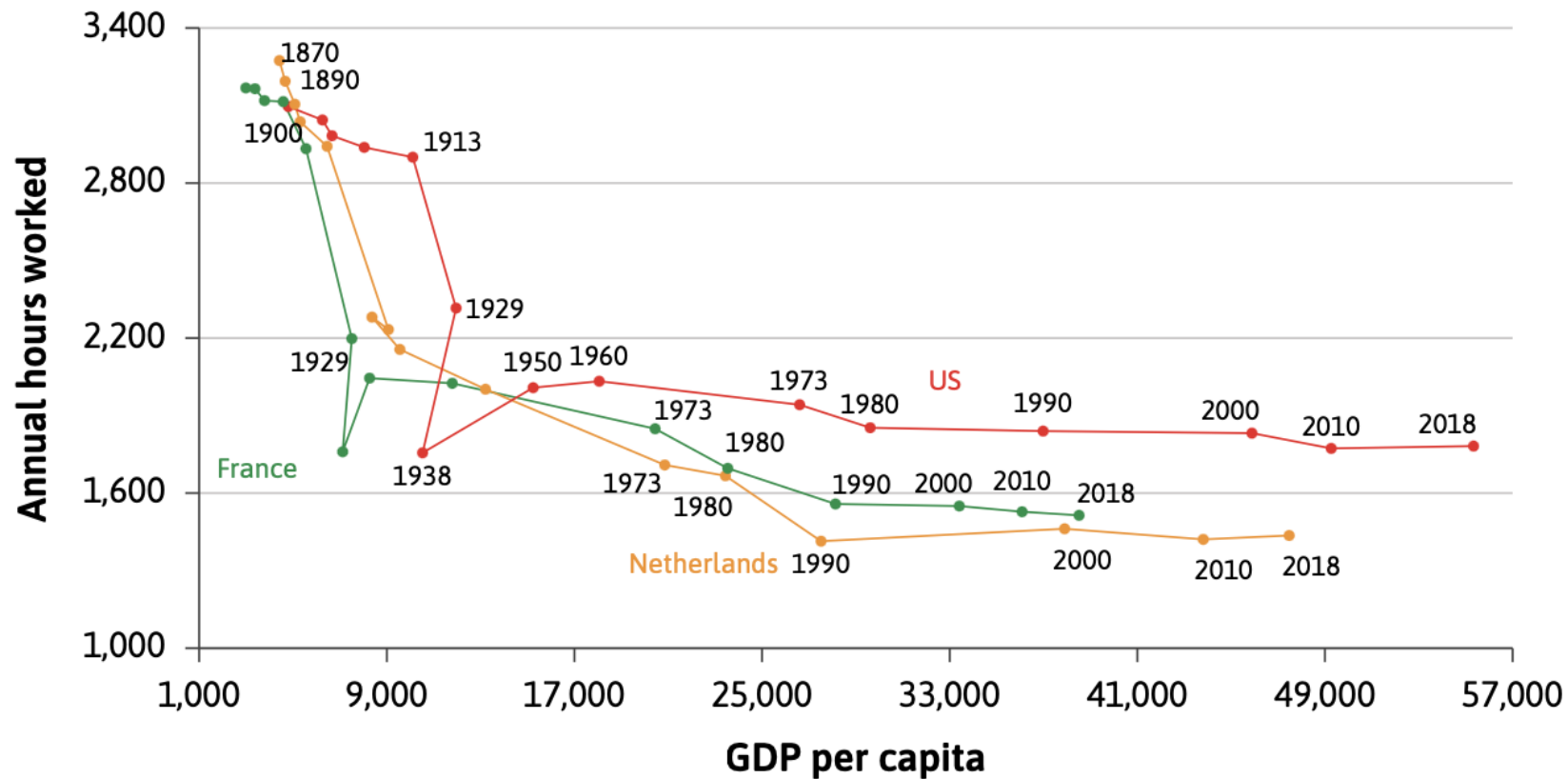
- A. Introduction
- B. Scarcity and choice
- C. Decision-making under scarcity
- D. Income and substitution effects
- E. Is it a good model?
- F. Applications to the real world

A. Introduction

The Context for This Unit

- Labour is work. (Unit 1)
- Labour is an input in the production of goods and services. (Unit 2)
- New technologies raise the productivity of labour.
 - How would that affect living standards?
 - How would that affect the free time and working hours chosen by individuals?

This Unit



For most countries, living standards have greatly increased since 1870. But there are disparities in free time and income across countries.

This Unit

Use a **model of individual choice** to explain the differences in working hours:

- over time
- because of changes in inequality
- across genders
- across countries

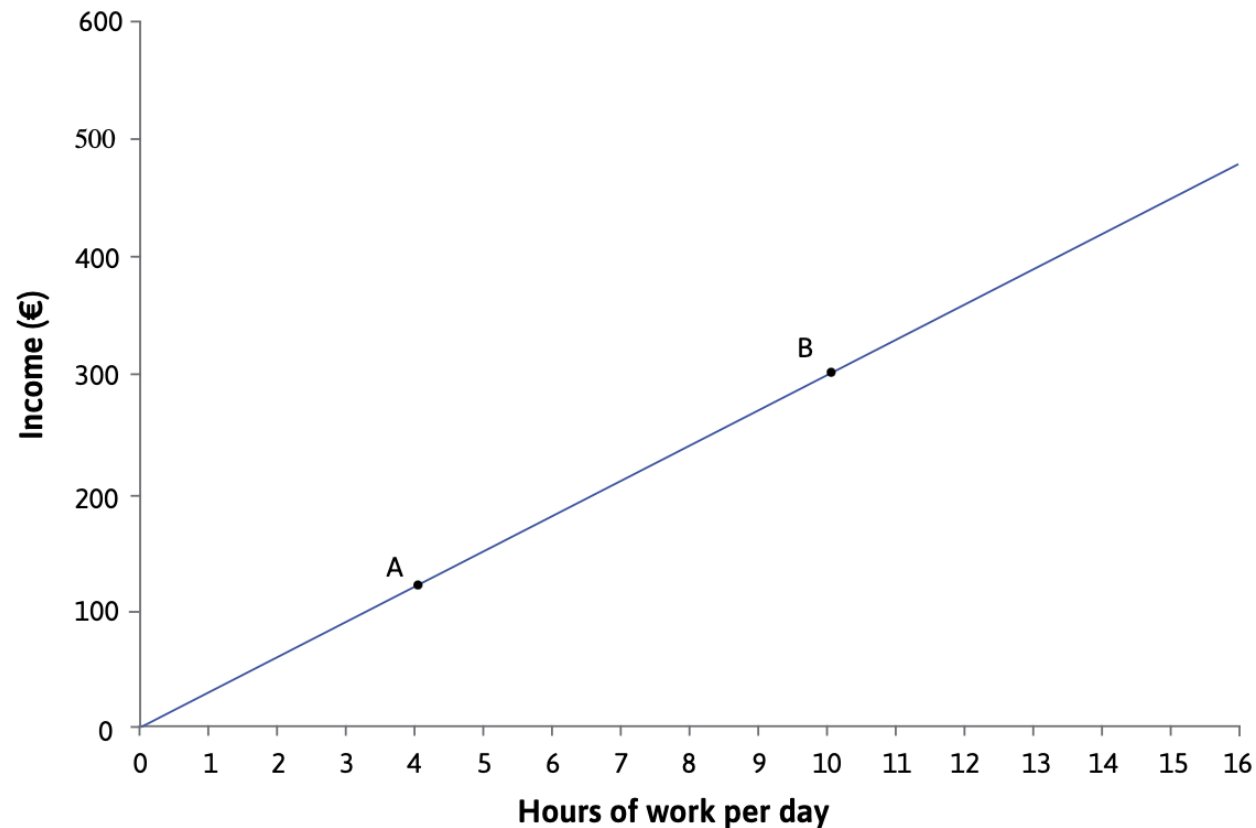
B. Scarcity and choice

A model of constrained choice

- Individuals want to consume as much as possible and have as much free time as possible.
- However, their consumption relies on working, and so sacrificing free time.
- How can we model how an individual chooses the combination of free time and consumption that is best for them?

Example: Karim moves to Madrid

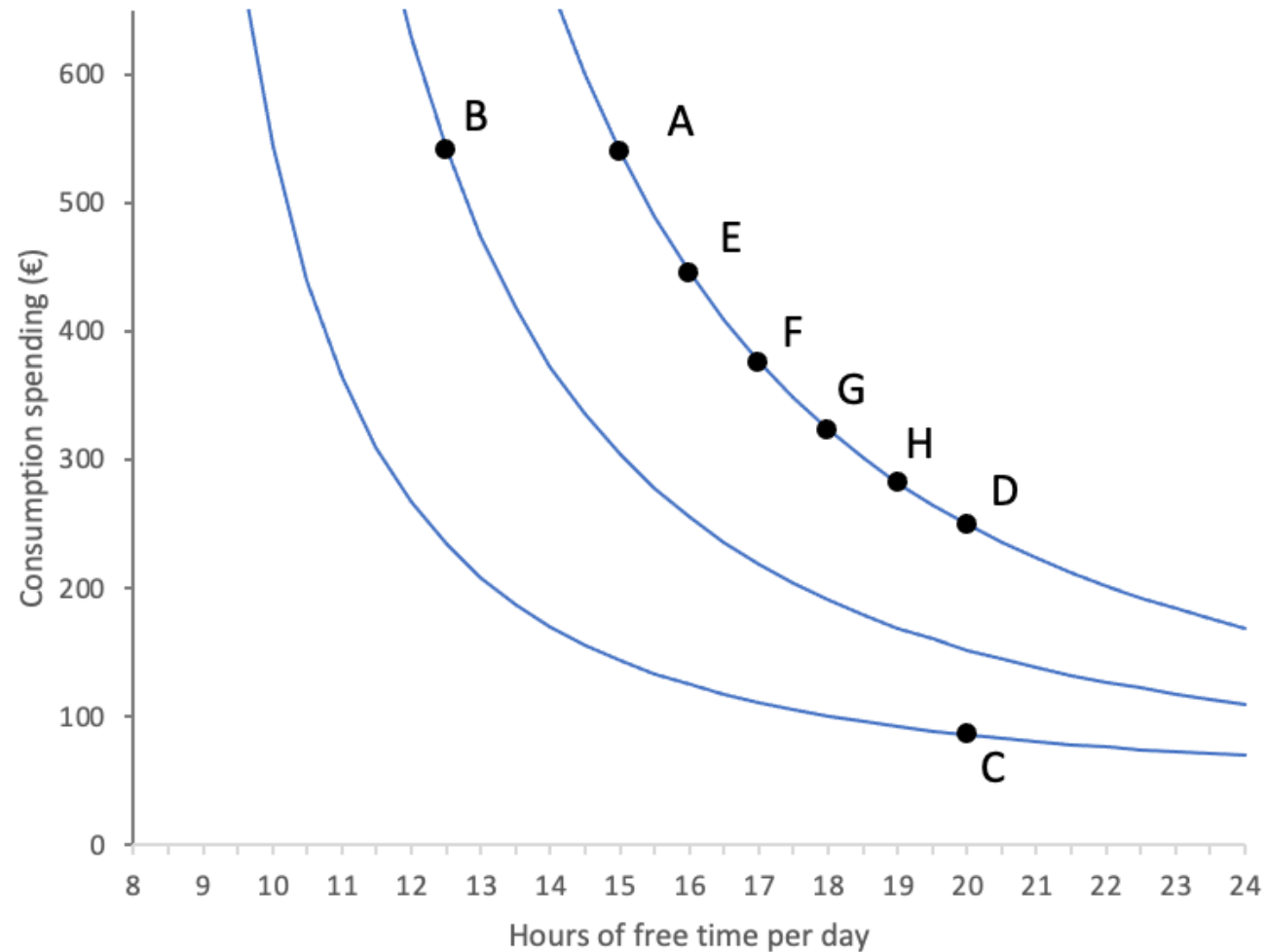
- Karim can earn €30 an hour and can work a maximum of 16 hours a day. How does Karim choose the number of hours to work?



Assumptions

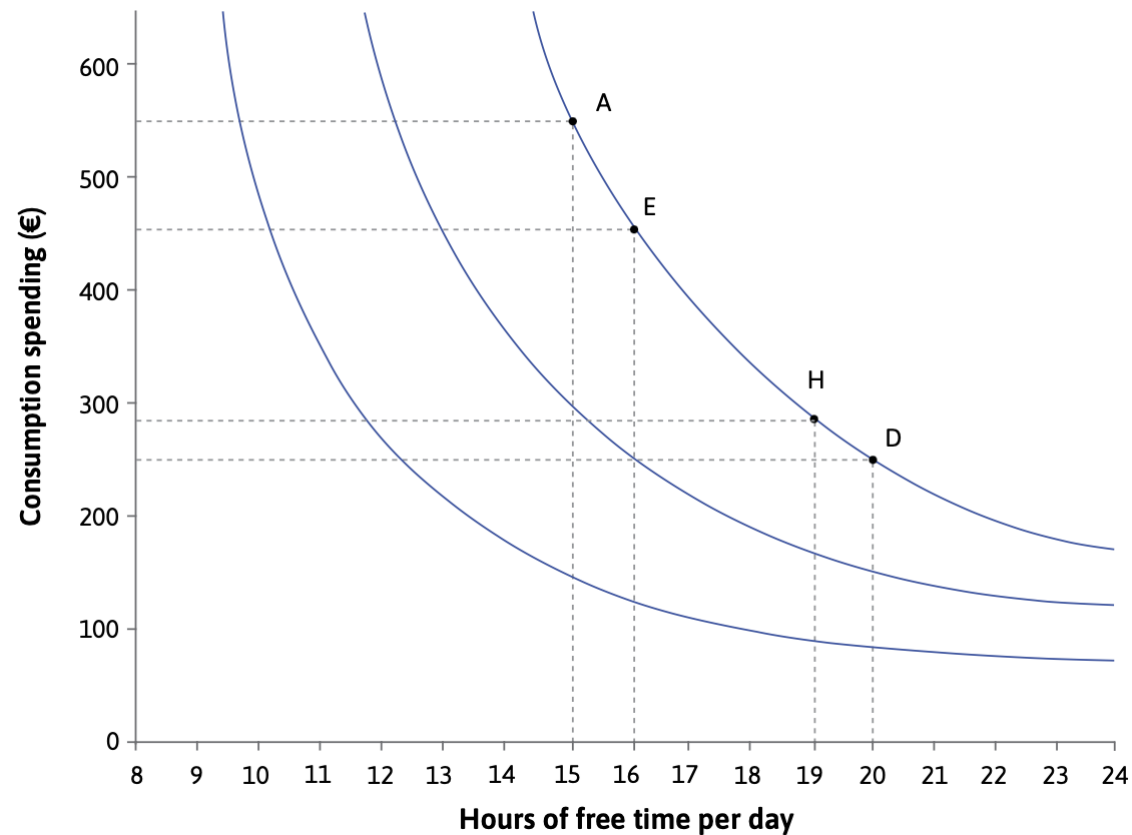
- Karim doesn't care about the future, so does not save any of his earnings.
- Karim cannot borrow so his average spending cannot exceed his earnings.

Preferences and indifference curves

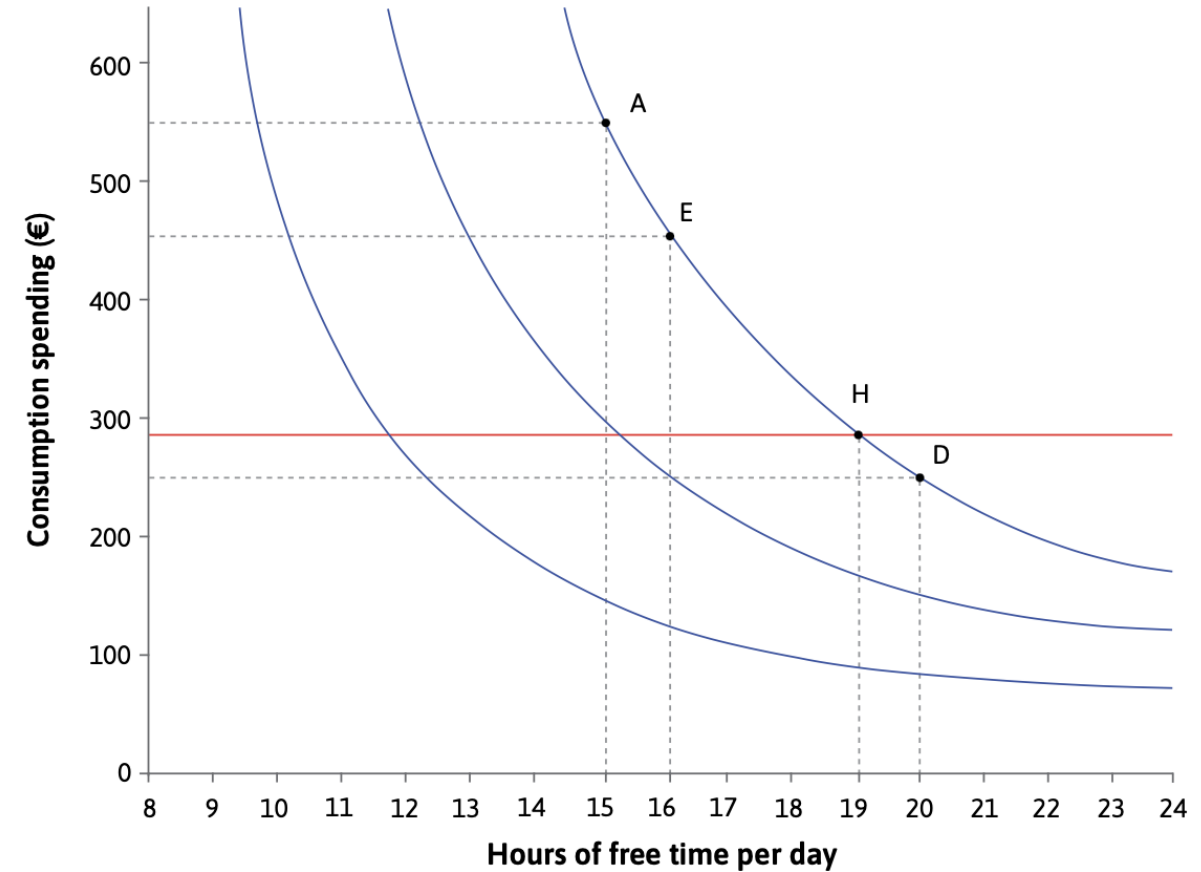
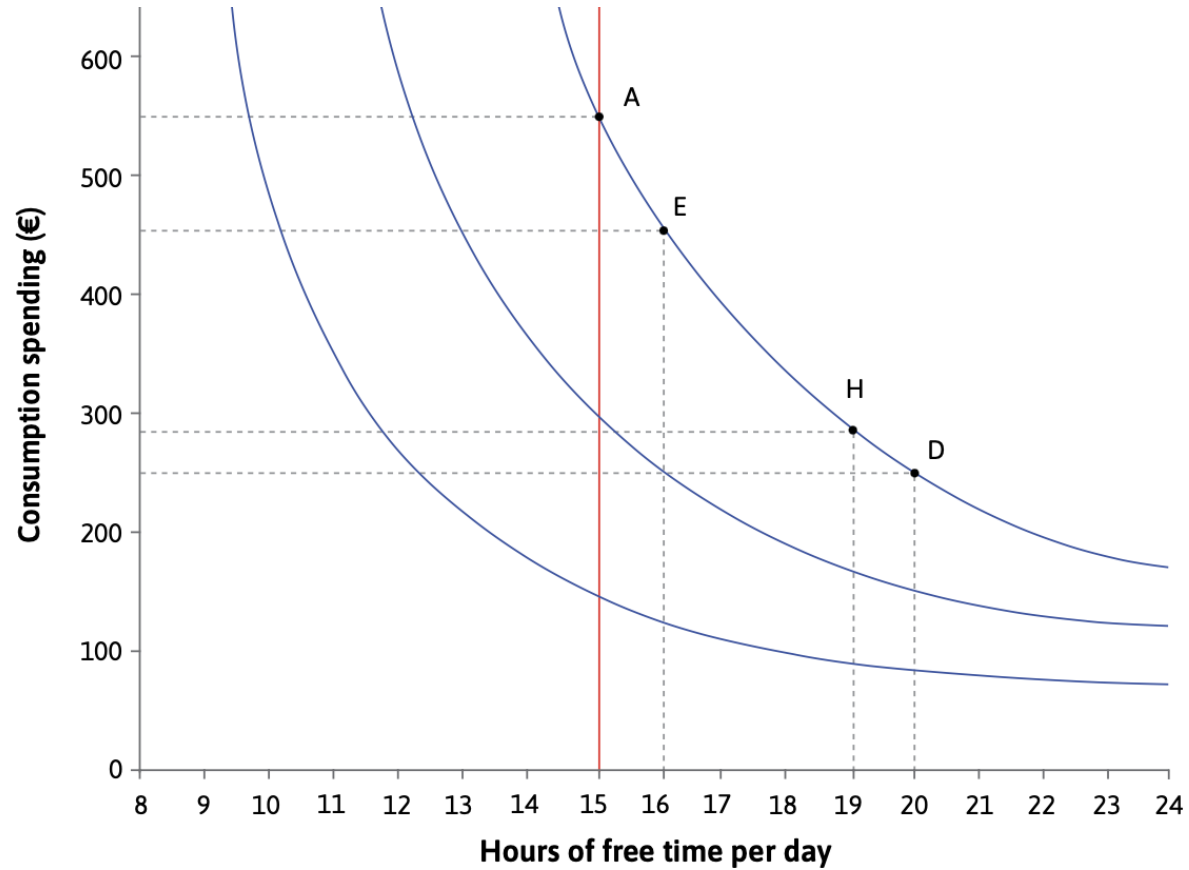


Marginal rate of substitution (MRS)

- The marginal rate of substitution measures the trade-off that Karim is willing to make between consumption and free time.

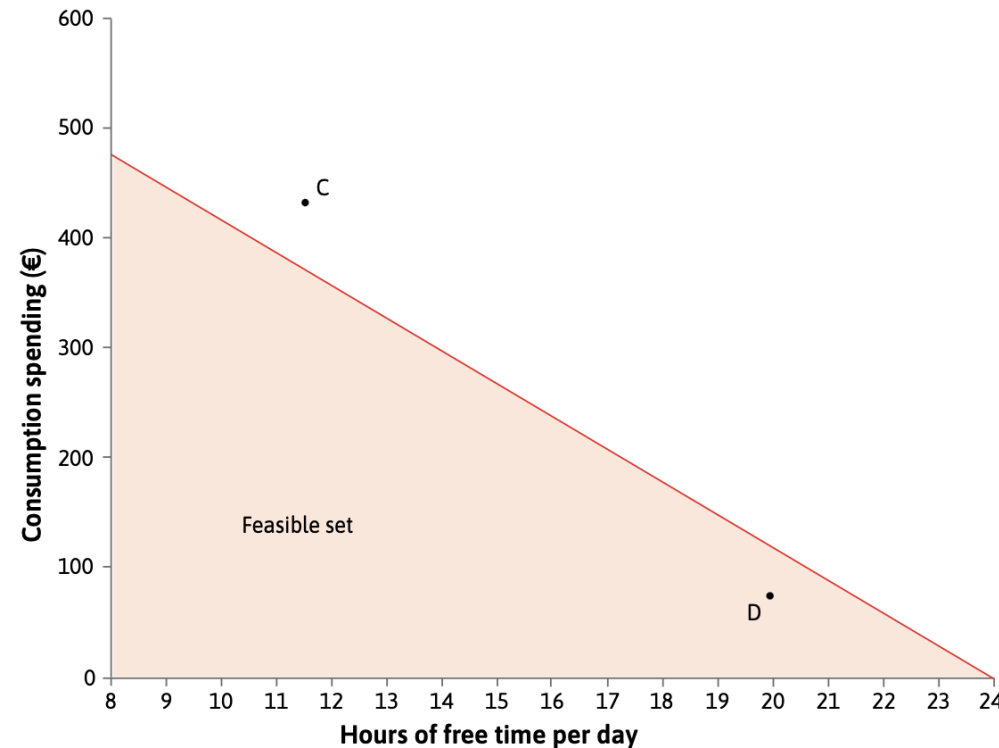


Marginal rate of substitution (MRS)



Opportunity cost, the feasible set

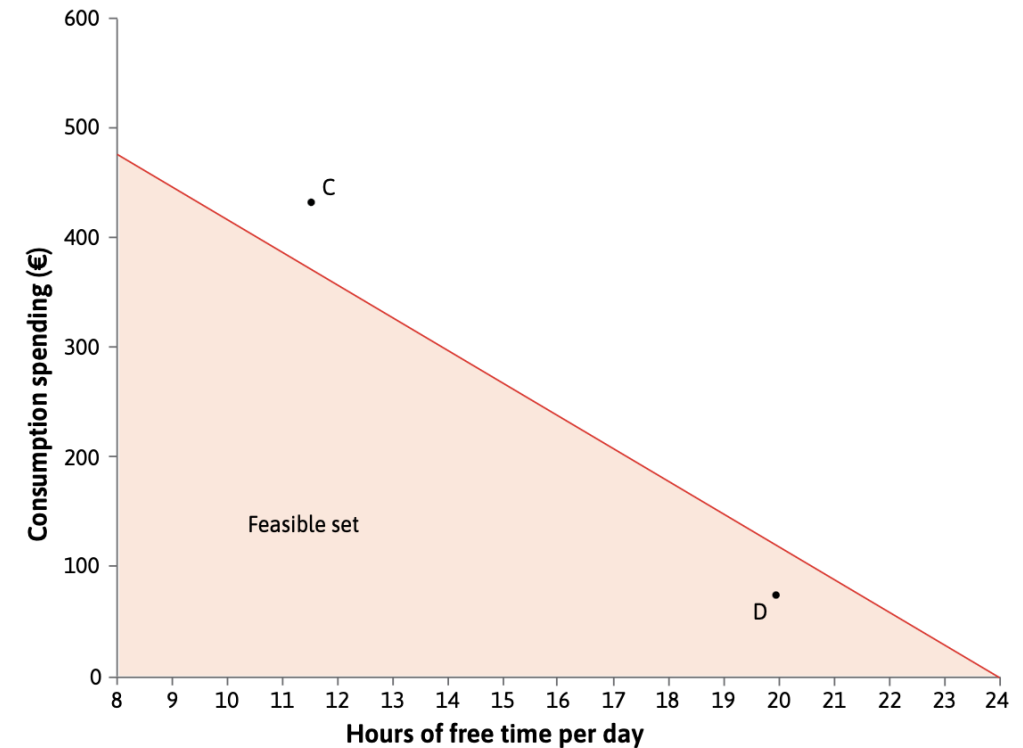
- Karim faces a dilemma: the more hours of free time he takes, the less consumption he can have. Free time has an **opportunity cost**.
- Karim's maximum level of consumption is given by: $c = w(24 - t)$



Marginal rate of transformation (MRT)

- The **marginal rate of transformation (MRT)** is the rate at which Karim can transform free time into consumption.

Hours of work	0	2	4	6	8	10	12	14	16
Free time (h)	24	22	20	18	16	14	12	10	8
Consumption (€)	0	60	120	180	240	300	360	420	480

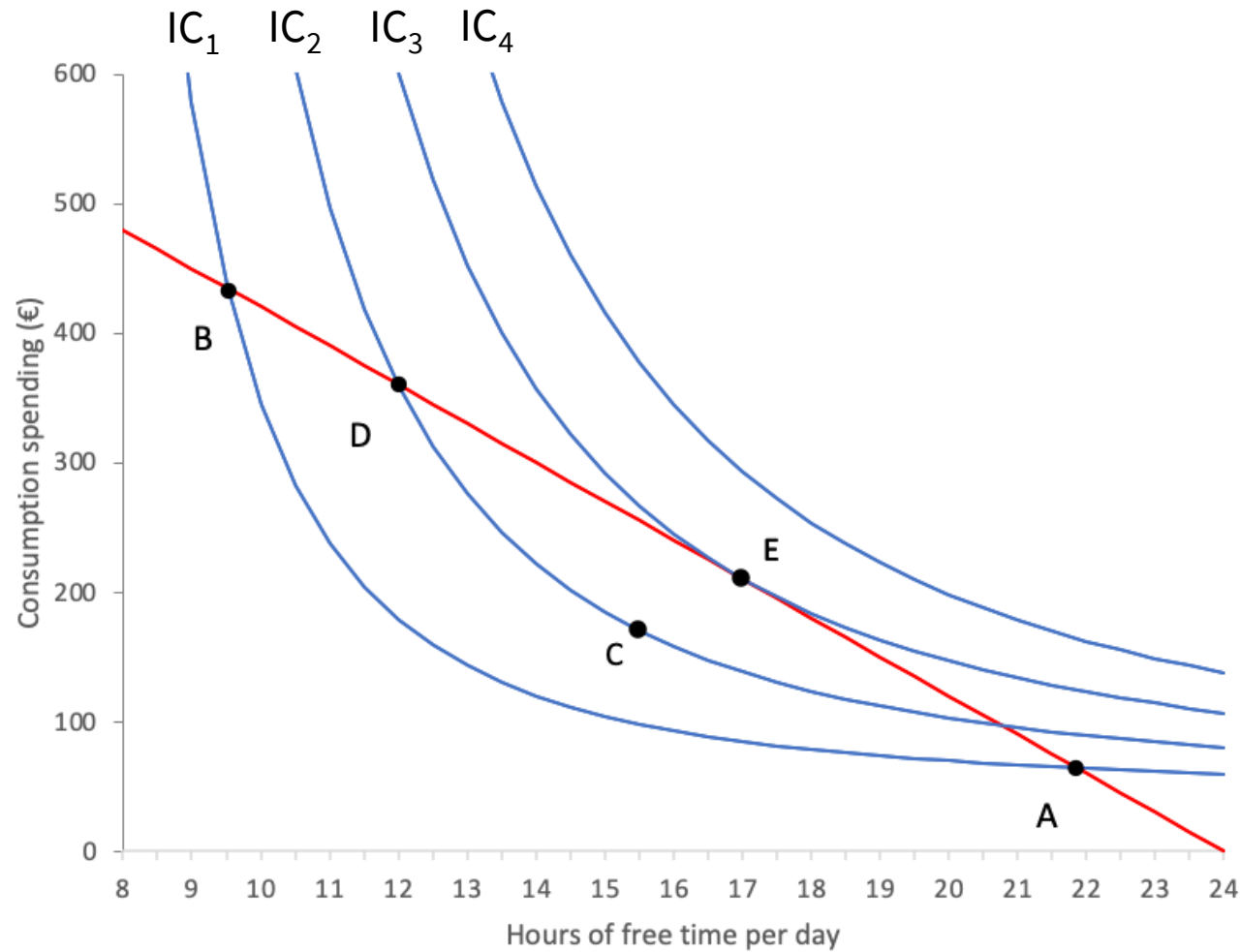


C. Decision-making under scarcity

The two trade-offs

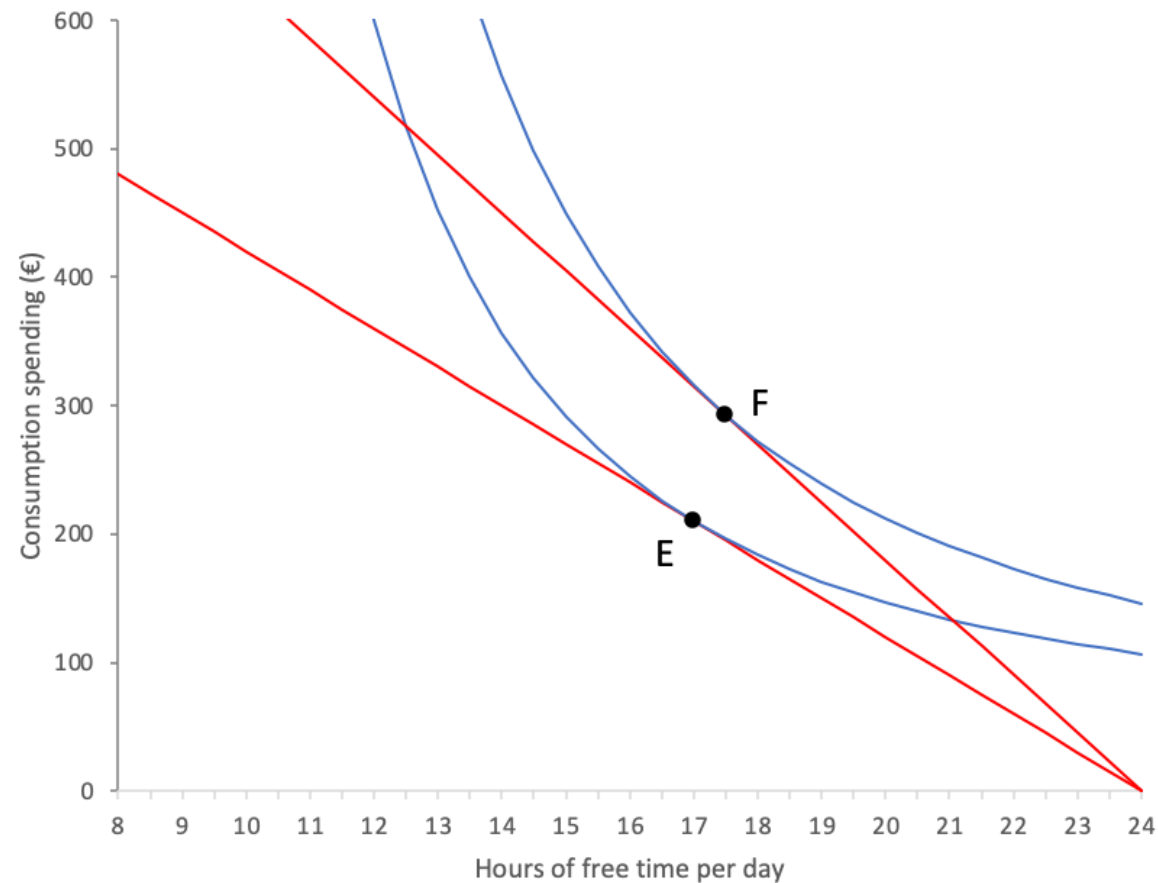
- The **marginal rate of substitution (MRS)** measures the trade-off that Karim is willing to make between consumption and free time.
- The **marginal rate of transformation (MRT)** measures the trade-off that Karim is constrained to make by the feasible frontier.

Decision making: $MRS = MRT$



Technological progress

- What if Karim's wage increases to €45?



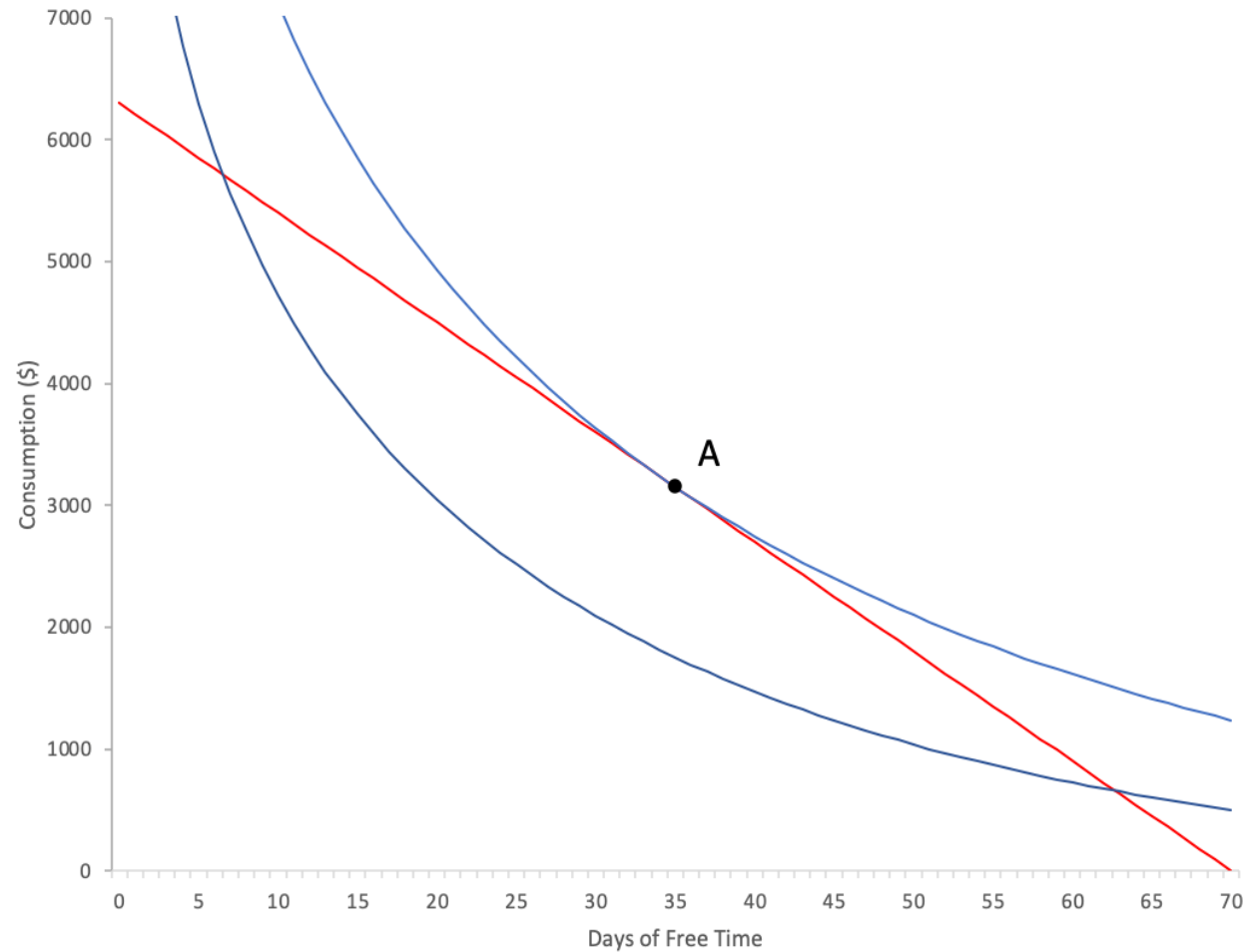
D. Income and substitution effects

A new scenario

- Imagine that you are planning how to spend the 10-week summer break before your next year at college. You have the opportunity to work in a local shop, where you would be paid \$90 per day. But you also want to have time to meet friends, take a holiday, and study for next year's courses. How many days should you work during the break?

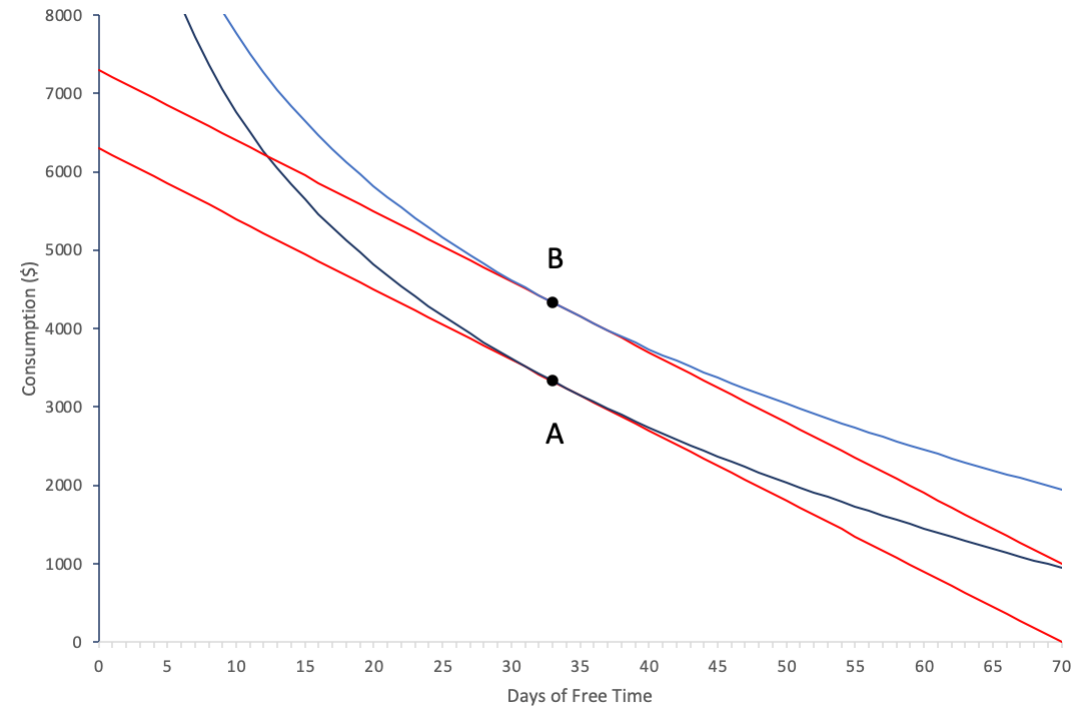
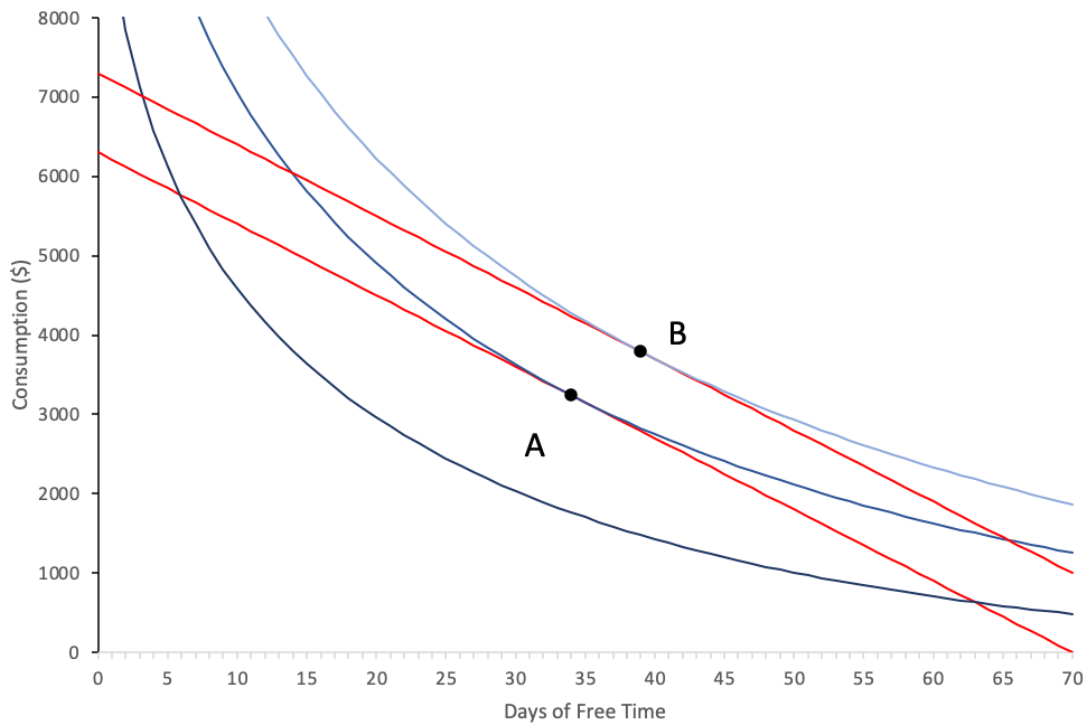
Making the best choice

- $MRS = MRT$



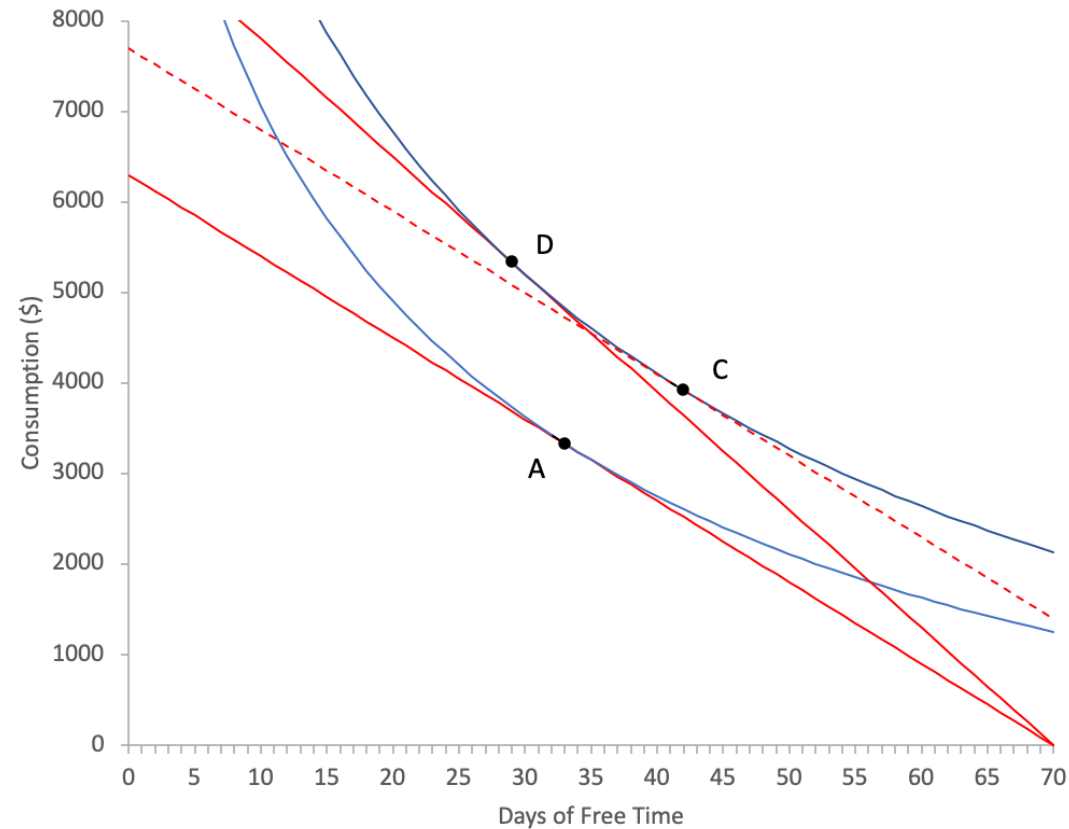
Extra income

- A mysterious benefactor would like to give you \$1000. How do you respond?



The superstore vacancy

- A new job is available at \$130 per day.



In summary

A wage rise has two effects on your choice of free time:

- The **income effect** (because the budget constraint shifts outwards): the effect that the additional income would have if there were no change in the opportunity cost.
- The **substitution effect** (because the slope of the budget constraint, the MRT, rises): the effect of the change in the opportunity cost, given the new level of utility.

E. Is it a good model?

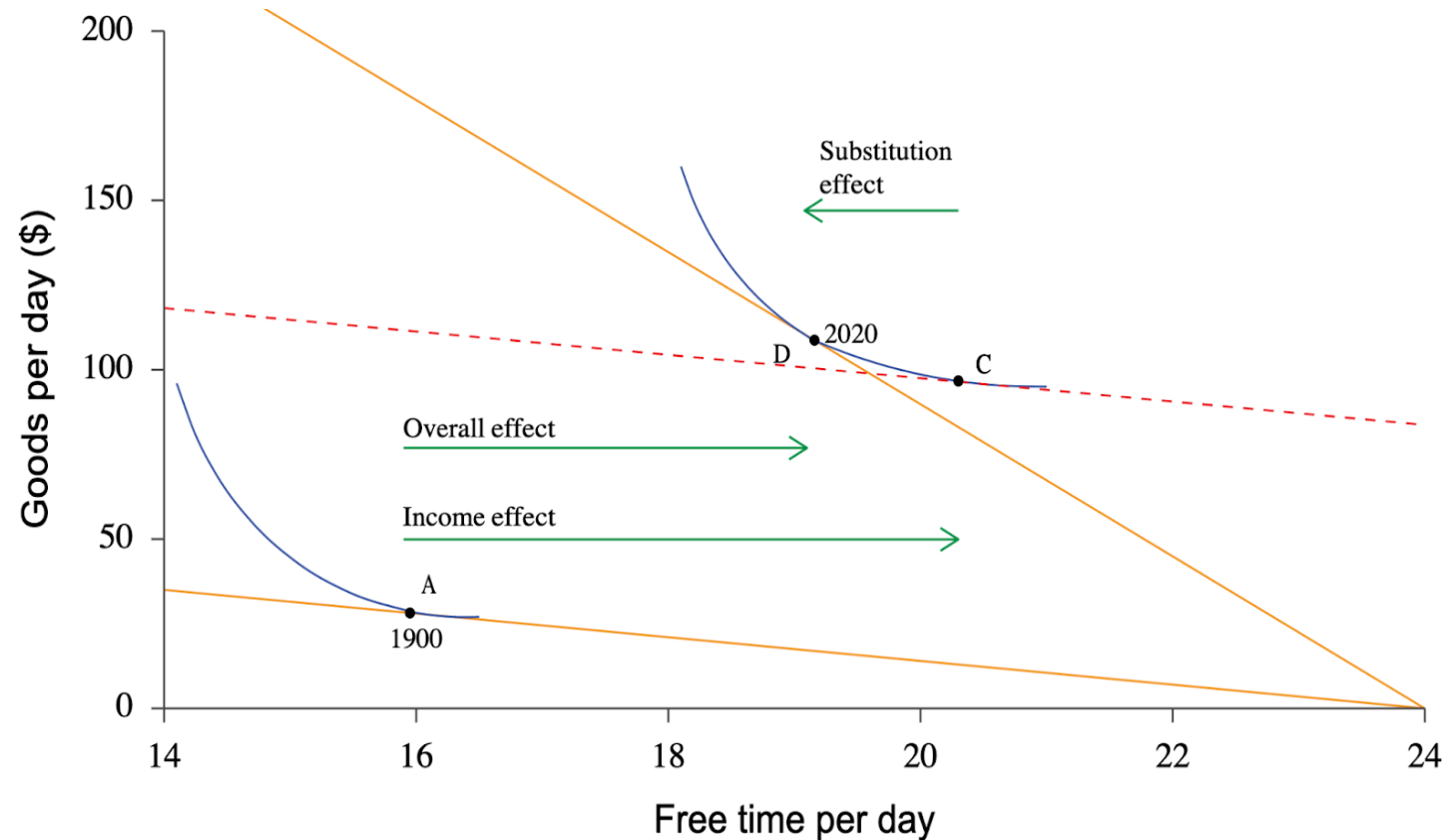
Problems and answers

- This is not what people do!
- What about other factors?
- What about large groups of workers?
- What about the influence of culture and politics?

F. Applications to the real world

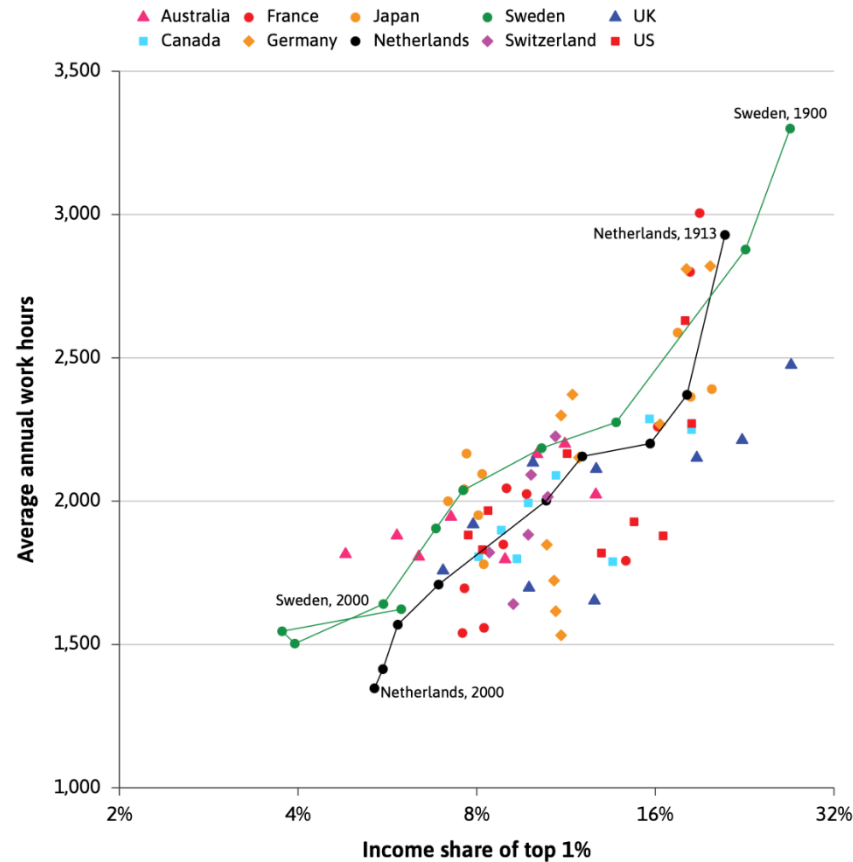
Changes over time

Consumption and free time per day in the US between 1900 and 2020



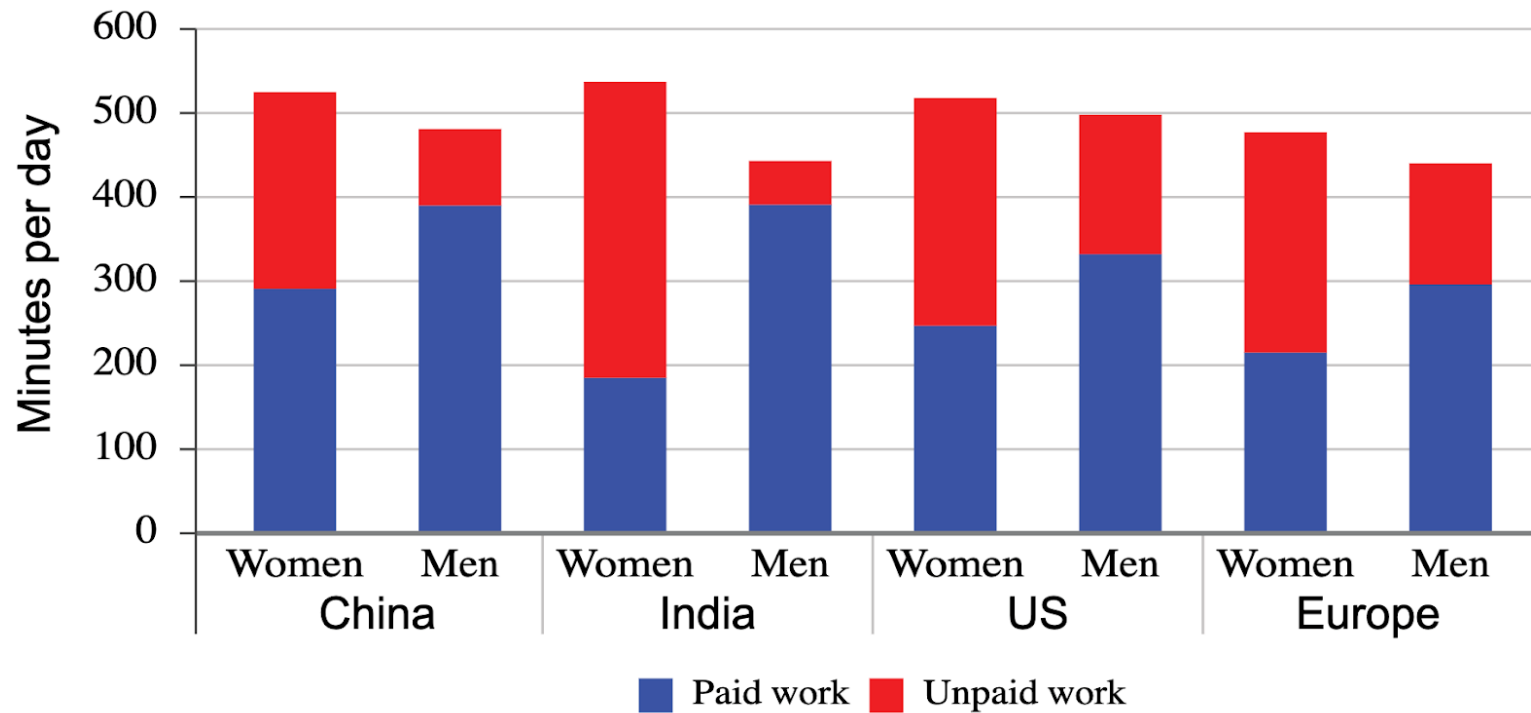
The effect of inequality

- Veblen effect or conspicuous consumption



The gender pay gap

- Paid vs unpaid work done by men and women



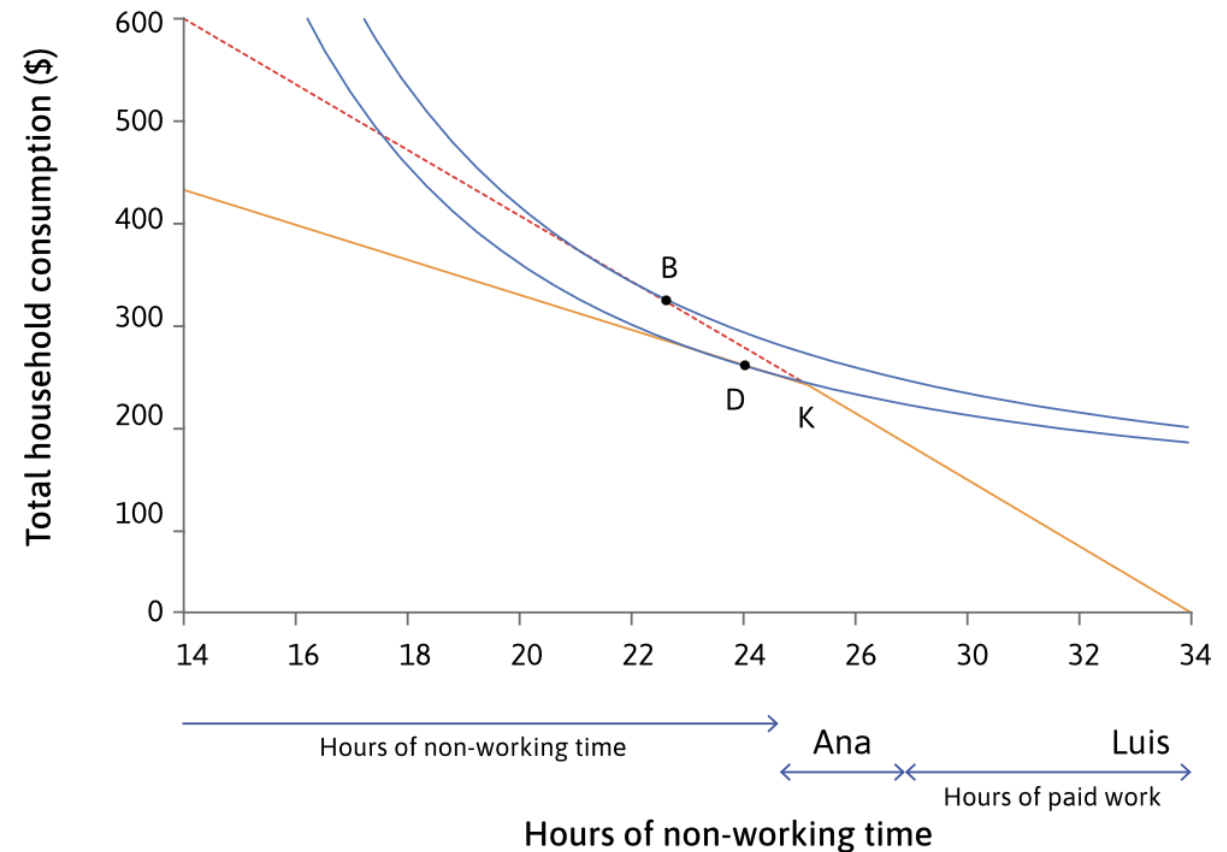
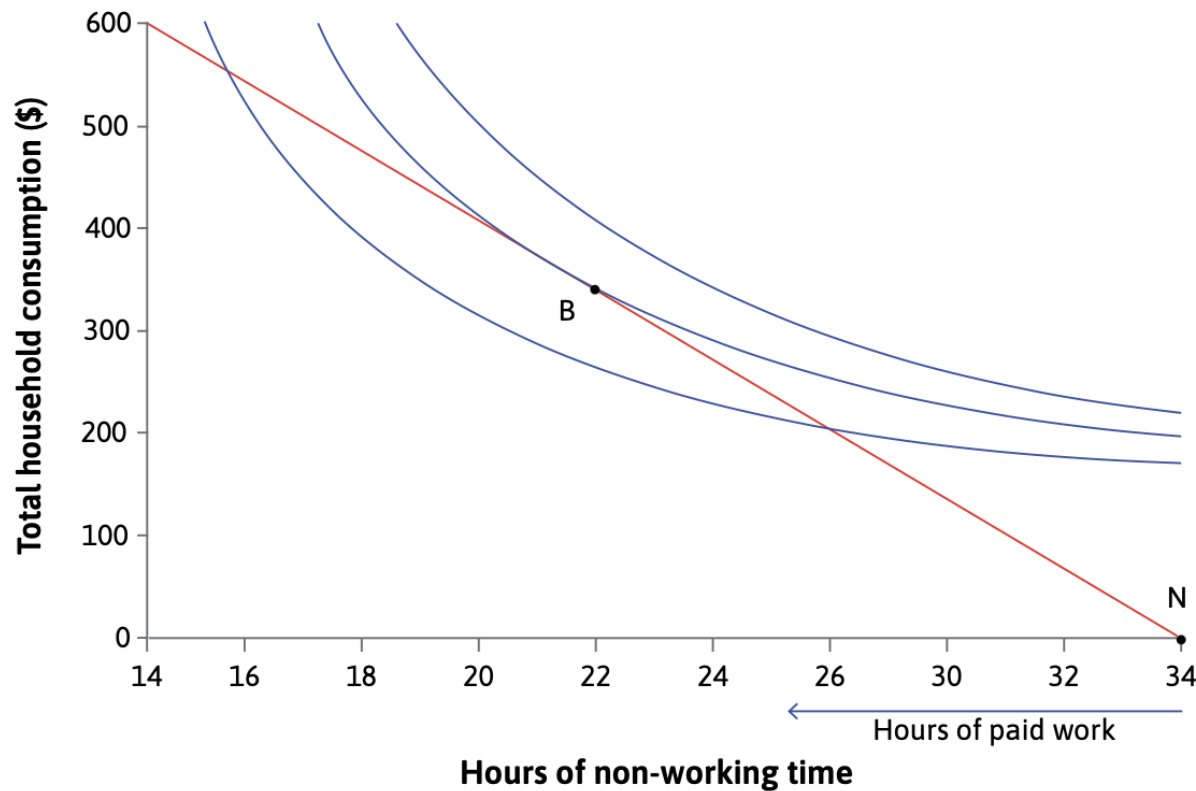
The gender pay gap

Consider Ana and Luis:

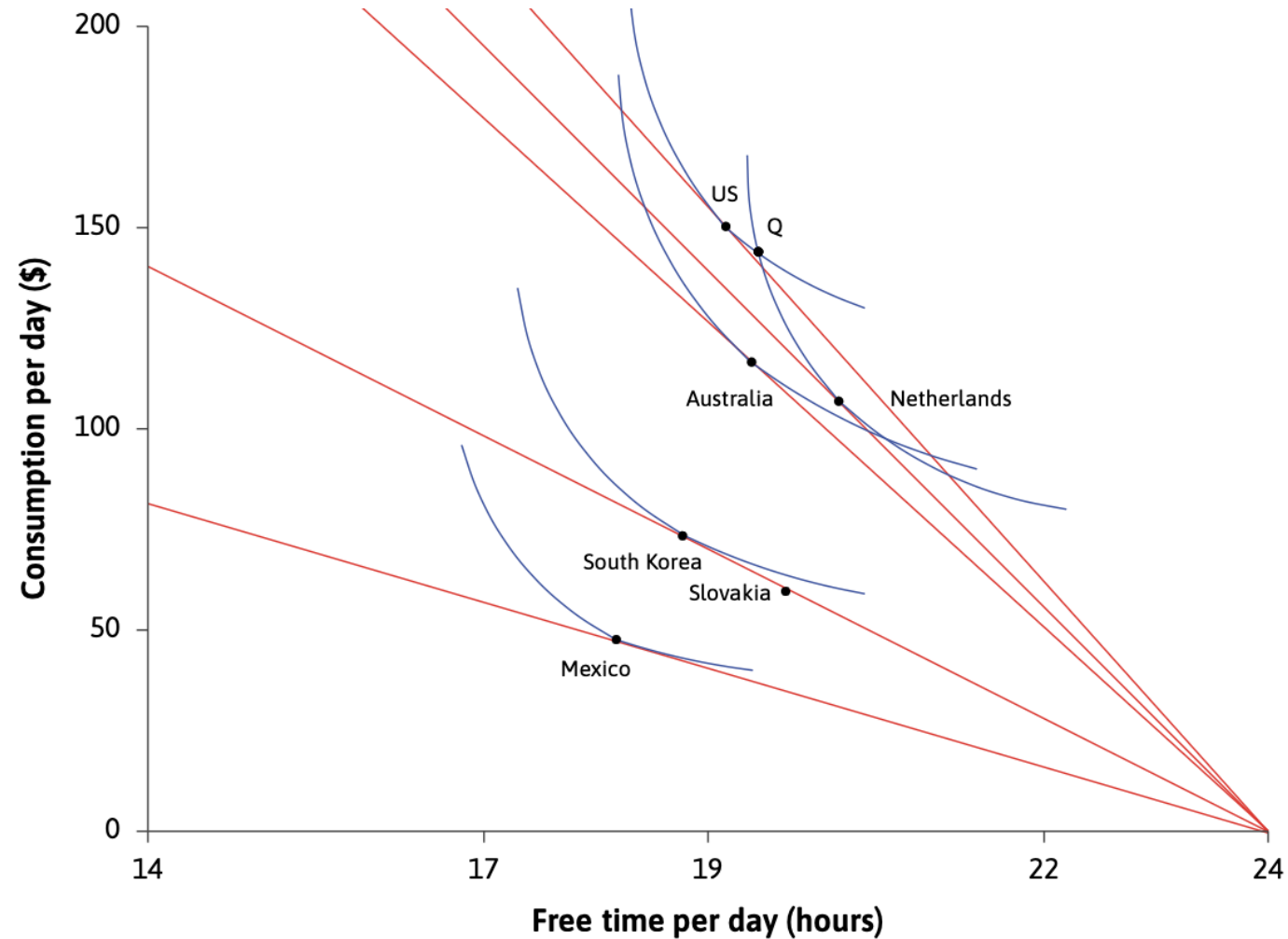
They are a couple with two young children. Suppose they have the same preferences and similar skills, and can both find employment at the same wage, \$30 per hour. They have worked out that the household needs 14 hours of domestic work, including childcare, per day.

The gender pay gap

- Ana now can only earn \$17 per hour. Luis is limited to 8 hours of work.



Differences across countries



Summary

1. Simple model of decision-making under scarcity
 - **Indifference curves** represent preferences
 - **Feasible frontier/budget constraint** represents choice constraints
 - Utility-maximising choice where **MRS = MRT**
2. Used model to explain effect of technological change on labour choices
 - Overall effect = **Income effect + Substitution effect**
 - Limitations of model – omits important factors

In the next unit

- Models of individual choice that include other important factors
- The role of **social interactions** in individual choice
- The effect of individual choice on **social outcomes**